PS #38 Community Microgrid — Team Briefing

Oct 4, 2026 · @Res

In one paragraph

We plan to build a small **12 V DC community power network** for three neighbouring "houses" that share one battery, trade solar power using prepaid RFID cards, and keep running when the "grid" fails. This doc explains what PS #38 asks for, how the system would work, the decisions still open, and where you fit in. Nothing is built yet; it is a concept you can question and improve.

Timing matters: per the rulebook, Round 1 (online PPT plus a circuit-simulation explanation) is on 5–6 Oct, and Round 2 (offline prototype at WCE Sangli) is on 8–9 Oct. Round 1 needs the simulation story more than finished hardware.

What PS #38 asks for

The official statement: *"Develop an intelligent microgrid controller that detects grid failure, manages locally generated renewable energy and storage, prioritizes critical loads, and safely manages the transition between grid-connected and islanded operation."* The competition portal adds the hardware idea: *a prepaid RFID-based energy dispensing meter that lets neighbouring households trade and route localised solar power.*

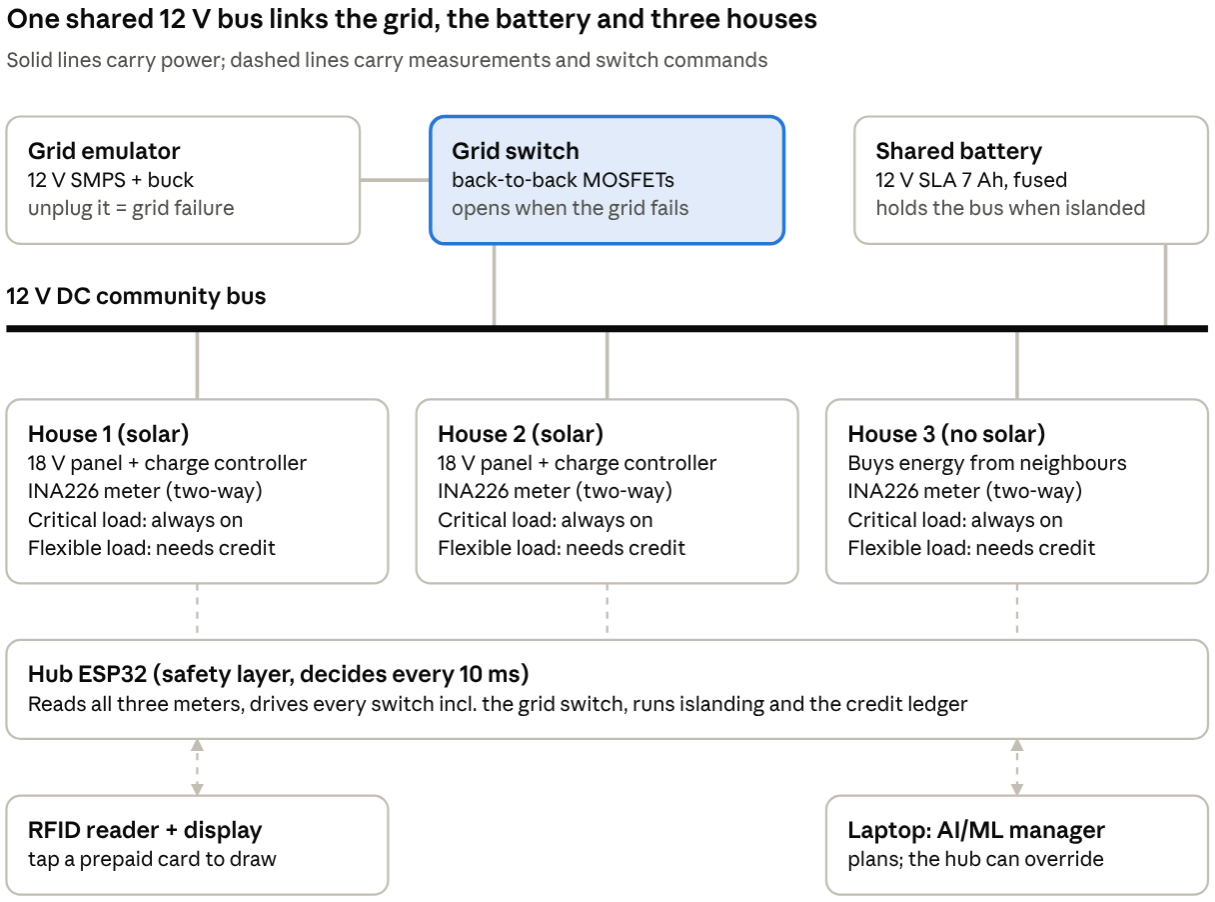
So the jury will look for five things working together:

1. **Grid-failure detection** — notice within a fraction of a second that the grid is gone.
2. **Islanding** — disconnect from the grid and keep the community running on solar and battery alone.
3. **Critical-load priority** — essential loads (a light, a medical device stand-in) stay on; others are shed first.
4. **Safe reconnection** — rejoin the grid when it returns, without a current surge.
5. **Prepaid metering and trading** — each house taps a card, buys energy with credit, and can sell its surplus solar to neighbours.

The jury also scores methodology, documentation, technical understanding and viva answers, so every team member must be able to explain the system.

The proposed system

Everything hangs off one 12 V DC bus: the grid emulator (through the grid switch), the shared battery, and three houses. One ESP32 hub measures every house and controls every switch.



proposed PS #38 system · one bus, three houses, one hub

When the grid switch opens, the battery and the two solar houses keep the bus alive on their own; that is islanding.

Why a 12 V DC bus instead of the suggested AC kit

The portal's suggested kit is built around 230 V AC: a PZEM-004T (a mains energy meter), 30 A relays and optocouplers. We are choosing a 12 V DC bus instead. The rules let us pick our own components, as long as we bring them.

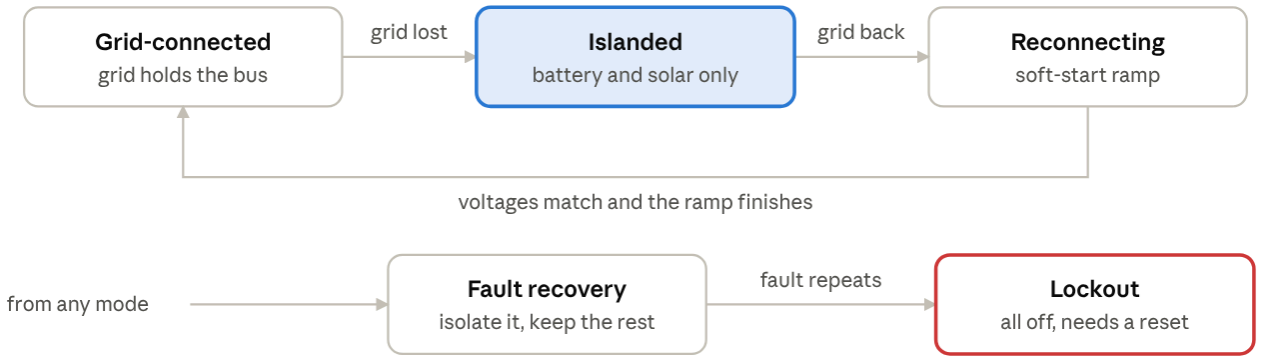
	Suggested kit (AC)	Our plan (12 V DC)
Safety on a bench	230 V mains on a breadboard is dangerous and venue rules restrict it	12 V is safe to touch
Solar and battery	Need an inverter to put them on AC	Connect almost directly; solar and batteries are natively DC
Selling power back (two-way flow)	PZEM measures only one direction well	INA226 sensor measures current both ways, with sign
Islanding demo	Needs AC sync (matching voltage, frequency, phase) before reconnecting	Only voltage needs to match; a soft-start ramp handles it
Switching	Relays: slow, click, wear out	MOSFETs: fast, silent, can be PWM-ramped
Realism	Matches Indian homes	Matches DC nanogrids used in rural electrification and solar homes

We keep the kit items that still fit: the **ESP32**, the **RC522 RFID module**, a **display** (16x2 I2C LCD or a small OLED), and the **12 V SMPS + buck**, which becomes our grid emulator. The PZEM and 30 A relays are skipped. In the pitch we should say this choice openly and explain that a full-scale version would add an inverter and AC metering on top of the same control logic.

How it works

The hub runs a **state machine** (a fixed set of modes, with clear rules for moving between them). The diagram below shows the modes; here is what each part of the behaviour means in practice.

The hub islands on grid loss and rejoins only through a soft ramp



hub operating modes · 3 main modes, 2 fault modes

Grid-connected (normal day). The grid emulator holds the bus near 12 V. Solar houses use their own solar first; surplus goes to neighbours, then to the shared battery. Each meter records energy in and out, and the hub moves credit between card wallets.

Grid failure and islanding. The hub watches bus voltage and grid current. If the grid disappears, it opens the grid switch within a fraction of a second, so our battery never tries to feed the outside world. The battery now holds the bus up. Low-priority loads are shed in a fixed order as the battery drains.

Reconnection. When the grid returns and stays stable for a set time, the hub checks that voltages match and closes the grid switch with the soft-start ramp (PWM: switching the MOSFET on and off very fast while slowly increasing the on-time), so there is no current spike.

Prepaid cards. Each house taps its RFID card to start drawing. Credit falls as it uses energy and rises when it sells solar. At zero credit, the house's flexible loads switch off. Three bad card taps lock that meter out.

Critical loads are a hard rule. Each house has one critical load that stays on even at zero credit and is the last thing shed in an outage. Its energy is still logged as debt.

Protection. Every source has a fuse. Software adds over-current trips, an I^2t check (a heating estimate, so brief inrush is allowed but sustained overload trips), over-voltage, and stuck-switch and frozen-sensor checks. A faulty house is isolated after two strikes without dropping the rest of the community. A physical **E-stop** cuts power to all MOSFET drivers, so every switch turns off.

How we will build it

We **simulate first, then build**. A computer model of the exact circuit (a "digital twin") runs every scenario before any part is wired: normal day, grid outage, low battery, bad card, faulty house, reconnect.

Every tunable number (voltages, current limits, timings, battery thresholds) lives in one parameters file, and the ESP32 firmware's constants are generated from it. We then run the real firmware code against the model to prove both behave the same (a SIL test). This is also the "circuit simulation" the Round 1 jury asks hardware teams to explain.

Open decisions

With Round 2 four days away, every recommendation below favours what we can build and debug by 8 Oct. Your input on the battery and the hardware scope is most useful.

#	Decision	Options	Recommendation and why
1	How power flows both ways	A: battery sits directly on the bus. B: two-way DC-DC converter. C: several sources sharing by droop control	**A.** Simplest and robust; B and C need a week of tuning we don't have.
2	Battery type	LiFePO4 4S + BMS, or sealed lead-acid (SLA)	**SLA 12 V 7 Ah** unless we already own a LiFePO4 pack: sold in any local electronics shop, tolerant, cheap. Must still be fused.
3	Houses	3 houses: two with solar, one without	**Keep 3.** House 3 having no solar is what makes trading visible.
4	Grid failure	Real grid emulator behind a switch, or a software flag	**Real emulator.** Pulling a plug in front of the jury is the strongest demo moment.
5	Hardware scope	One ESP32 per house + hub (radio link), or one ESP32 for everything	**One ESP32 first**, with three INA226 sensors on one I2C bus and one card reader. Split into per-house meters only if time allows.

#	Decision	Options	Recommendation and why
6	Solar at the venue	Real panels, or a lamp / bench supply as a "PV emulator"	**Bring both.** Indoor light may not drive an 18 V panel; a current-limited supply keeps the demo repeatable.
7	Where the AI/ML energy manager runs	Hub, laptop, or cloud	**Laptop**, talking to the hub over USB serial or Wi-Fi. The hub keeps the final say on safety.
8	Team name	Shortlist: AMRIT, Govardhan Grid, Mohalla Grid, Saanjhi Urja, UrjaMesh	Team's call; quick web checks only, not trademark searches.

Estimated source current at the 12 V bus is not yet worked out; the params file will fix fuse ratings and battery size together.

What data the system will produce

The simulator and the real hub will log the same columns, one row per time step, so the AI/ML energy manager can be built on simulated data and then run on hardware unchanged. This is a draft; units are SI (W, V, A, Wh), power positive into the bus.

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t_s,mode,grid_ok,v_bus,p_grid_w,p_batt_w,soc,h1_pv_w,h1_crit_w,h1_flex_w,h1_credit,h2_pv_w,h2_crit_w,h2_flex_w,h2_credit,h3_pv_w,h3_crit_w,h3_flex_w,h3_credit,fault
0,GRID,1,12.6,4.1,-2.0,0.62,6.5,1.0,2.4,50.0,5.8,1.0,0.0,42.0,0.0,1.0,3.0,30.0,NONE
1,ISLAND,0,12.4,0.0,3.6,0.62,6.5,1.0,0.0,50.0,5.8,1.0,0.0,42.0,0.0,1.0,0.0,30.0,NONE
```

The manager sends back a small command (battery power set-point, which flexible loads to shed, trade price). The hub may refuse any command that breaks a safety limit. The full field list and the AI/ML side are in the separate AI/ML brief. The numbers in the sample rows are illustrative, not simulated.

Safety rules and plan

Non-negotiable bench rules:

- 12 V DC only; nothing on the bench touches mains except the SMPS's own sealed input.

- A fuse on every source: battery, grid emulator, each solar input. A lead-acid battery can deliver hundreds of amps into a short and melt wires.
- Sources are switched in the + wire with P-MOSFETs (high-side); loads in the − wire with N-MOSFETs (low-side). All grounds are shared, which is why sources can't be switched in the ground wire.
- Every switch is OFF at power-up (pull-down/pull-up resistors guarantee it).
- The E-stop cuts the MOSFET drivers' supply, not the logic, so one press turns every switch off.
- Check battery polarity twice before connecting; reverse polarity destroys the ESP32 and sensors.

Plan to Round 2:

1. 4–5 Oct: lock decisions, write the params file, build the digital twin and run the six scenarios (normal day, grid outage, low battery, bad card, faulty house, reconnect). This is the "circuit simulation" Round 1 asks for.
2. 5–6 Oct: Round 1 pitch with diagrams and simulation plots.
3. 6–7 Oct: buy parts, bench bring-up in phases: one house meter, then battery, then grid switch, then full island test.
4. 8–9 Oct: Round 2 demo: normal trading, pull the grid plug, critical loads stay on, reconnect smoothly, a bad card is refused.

Glossary

Term	Meaning here
Bus	The shared + and − wires every source and load connects to
Islanding	Running the community on its own after disconnecting from the grid
Critical load	A load that must never go off (stand-in for a medical device or emergency light)
Shedding	Switching off low-priority loads to save energy
SOC	State of charge: how full the battery is, 0 to 1
Soft-start	Ramping a connection up gradually with PWM to avoid a current spike

Term	Meaning here
PWM	Pulse-width modulation: switching very fast and varying the on-time to control average power
High-side / low-side switch	A switch in the + wire / in the – wire
Back-to-back MOSFETs	Two MOSFETs in series facing opposite ways, so the switch blocks current in both directions when off
INA226	A small I2C sensor that measures bus voltage and current in both directions
ESP-NOW	A direct ESP32-to-ESP32 radio link that needs no Wi-Fi router
State machine (FSM)	Controller logic built as named modes plus rules for switching between them
Digital twin	A simulation of our exact hardware, used to test the logic before building
SIL test	Software-in-the-loop: running the real firmware code against the simulation to prove they match
I^2t	A running estimate of wire/part heating from current, used to trip on sustained overload